VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS ANALYSIS

CONTINUOUS AMBIENT AIR QUALITY MONITORING

Ambient CO Gas Analyzer *NDIR Gas Filter Correlation*

Carbon monoxide measured with a long-path curved mirror cell and a durable gas filter correlation wheel — with automatic zero correction and a year of logged data on board.



50 · 100 ppm
USER-SELECTABLE RANGES

7 inch
TFT LCD TOUCHSCREEN

1 year
ON-BOARD DATA LOGGING

NDIR GFC REFERENCE METHOD | **LONG-PATH MIRROR** OPTICAL CELL | **CPCB** ALIGNED REPORTING

50ppb
LOWER DETECTABLE LIMIT

2
SELECTABLE MEASURING RANGES

60s
TYPICAL RESPONSE TIME T90

32bit
ARM PROCESSOR, LINUX BASED

01 OVERVIEW

Carbon monoxide, *read continuously.*

A continuous ambient CO analyzer built on non-dispersive infrared gas filter correlation (NDIR GFC) — the accepted technique for regulatory carbon monoxide measurement. A curved optical mirror gives a long light path in a compact cell, and a specially designed, no-leak GFC wheel keeps the correlation signal stable over a long service life.

The instrument runs a 32-bit ARM processor on a Linux kernel with a 7-inch touchscreen interface, an automatic zero correction system for unattended stability, and Ethernet, RS-232, RS-485, USB and analog interfaces so the analyzer reports into a station, a SCADA system and a regulatory server at the same time.

REPORTED CHANNELS

- CO INSTANT
- 1-MIN AVERAGE
- HOURLY AVERAGE
- 8-HR AVERAGE
- 24-HR AVERAGE
- SAMPLE FLOW
- CELL TEMPERATURE
- IR SOURCE INTENSITY
- ZERO & SPAN STATUS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Stable Infrared Source

An IR source selected for stable output keeps the correlation signal steady between calibrations and extends service intervals.

No-Leak GFC Wheel

A specially designed gas filter correlation wheel runs leak-free for a long service life, keeping reference and measure signals clean.

Touchscreen Operation

A 7-inch TFT LCD touch interface puts live values, trends, ranges and diagnostics in front of the operator.

One Year of Data On Board

An internal NAND-flash logger holds up to a year of one-minute averages, downloadable over USB or network.

02 MEASUREMENT PRINCIPLE

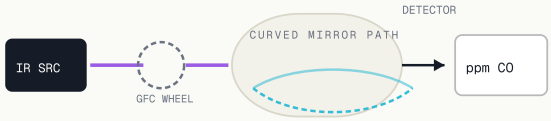
How NDIR GFC *counts CO.*

Carbon monoxide absorbs infrared light at a specific wavelength. Comparing absorption through a spinning gas filter correlation wheel against a reference gives a signal proportional to concentration — a direct optical measurement with no consumables in the sample path.

NDIR

Long-Path Absorption

IR SOURCE · CURVED MIRROR CELL · DETECTOR



Infrared light travels a folded, long optical path through a curved-mirror cell holding the sample. CO absorbs a characteristic wavelength; the detector reads the resulting attenuation.

`Absorption ∝ CO concentration`

CORRELATION

Gas Filter Correlation Wheel

SPINNING REFERENCE / MEASURE FILTER



A spinning wheel alternates a CO-filled reference cell and an empty measure cell across the beam. The differential between the two signals is a self-referencing measurement, immune to slow drift in the source.

`Differential signal ∝ CO`

STABILITY

Auto Zero & Compensation

AUTOMATIC BASELINE · T AND P CORRECTION



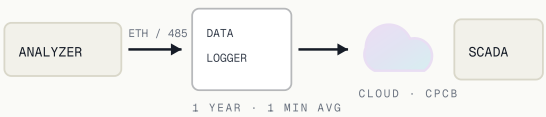
An automatic zero correction cycle re-establishes the baseline without operator intervention, while temperature and pressure compensation holds the reading steady as ambient conditions move through the day.

`Unattended long-term operation`

DATA PATH

Logging & Reporting

ETHERNET · RS-232 / 485 · USB · ANALOG



Readings are stored internally as one-minute averages for up to a year and transmitted live to a station data logger, plant SCADA or a pollution control board server.

`Download over USB or network`

03 SAMPLE PATH

From inlet to *reading.*

Everything between the sample line and the display is designed to keep the optical path clean and the correlation wheel turning true, run after run.

STEP 01

Filtered intake

Ambient air is drawn through a particulate filter and inert sample line into the analyzer at a controlled flow.

STEP 02

Long-path cell

Sample fills the curved-mirror optical cell, giving a long absorption path for good sensitivity in a compact housing.

STEP 03

GFC wheel

The spinning gas filter correlation wheel alternates reference and measure signals across the same beam and cell.

STEP 04

Detect and report

The differential is corrected for temperature and pressure, then displayed, logged and transmitted.

WHY THE WHEEL MATTERS

Gas filter correlation is self-referencing by design: the reference and measure signals pass through the same optics, cancelling out slow drift in the source or window fouling that would otherwise bias a single-beam reading. A no-leak wheel construction keeps that reference cell sealed for the life of the instrument, which is what makes low-ppm CO measurement repeatable rather than theoretical.

RANGE SELECTION

Two ranges — 0–50 and 0–100 ppm — are user selectable from the touchscreen. Urban background and CAAQMS sites usually run on the lower range for resolution; traffic corridors and near-source installations use the wider range so episodes are captured without saturating.

WHY CARBON MONOXIDE IS MONITORED

CO is a colourless, odourless gas produced by incomplete combustion — vehicle exhaust, industrial furnaces and biomass burning are the principal ambient sources. It binds haemoglobin in place of oxygen, so even moderate concentrations reduce the blood's oxygen-carrying capacity and aggravate cardiovascular conditions. Continuous measurement is what turns that risk into something a city or a regulator can act on.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

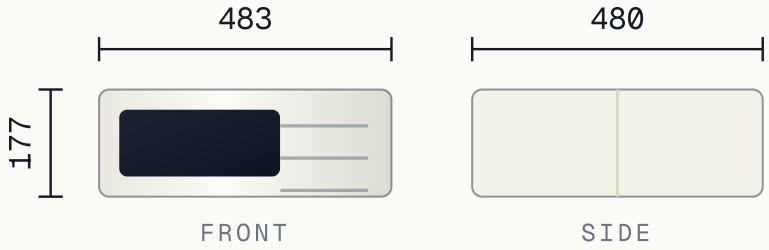
Parameter	Ambient carbon monoxide, CO
Working Principle	NDIR gas filter correlation
Measuring Ranges	0 - 50 / 100 ppm, user selectable
Lower Detectable Limit	50 ppb
Precision	±1 % of reading
Linearity	±1 % of full scale
Zero Drift	< ±1 % FS per 24 hours
Span Drift	< ±1 % FS per 7 days
Response Time	T90 typically 60 s
Optical Cell	Curved-mirror, long optical path
Correlation Wheel	Specially designed, no-leak construction

SYSTEM · COMMS · BUILD

Display	7-inch TFT LCD with touch interface
Controller	32-bit ARM Cortex-M4, Linux based multiprocessing
Zero Correction	Automatic · internal zero optional
Compensation	Temperature and pressure
Data Logger	NAND flash · up to 1 year of 1-minute averages
Interfaces	Ethernet · RS-232 · RS-485 · USB · analog out
Data Download	USB export or network transfer
Power Supply	230 VAC · 50 Hz
Mounting	19-inch rack or bench, station cabinet
Operating Conditions	5 - 40 °C, non-condensing
Enclosure	Weather-protected station cabinet for outdoor sites

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running.*

The analyzer is installed in a station cabinet with the sample manifold, calibration source and data logger. Per CPCB practice, the sample inlet sits in free-flowing ambient air **3 to 10 m above ground**, with an inert, short sample line and a scheduled **zero and span check** traceable to certified gas.

01 · MOUNTING

19-inch rack or bench

Standard rack fitment alongside other CAAQMS analyzers, sharing the station's manifold, power and data logger.

02 · SAMPLE LINE

Inert and short

PTFE line with a particulate filter, kept short and clean so CO reaches the optical cell without loss.

03 · ZERO AIR

Clean baseline source

A zero air unit supplies scrubbed air for the automatic zero cycle and for periodic manual verification.

04 · CALIBRATION

Multipoint traceable

Span with certified CO gas from a cylinder via a multipoint calibrator, on the site's audit schedule.

05 · CONSUMABLES

Filter and optics

Routine upkeep is the inlet filter, periodic leak check and mirror/optical cell inspection. No reagents and no wet chemistry.

06 · DIAGNOSTICS

Reported with the data

IR source intensity, cell temperature and pressure, flow and zero/span status travel alongside every reading.

PART OF A FULL STATION

The CO analyzer is normally specified with SO_2, NOx and ozone analyzers, a multipoint calibrator and zero air unit, a particulate monitor and a meteorological mast. Sharing a time base, cabinet and data path, they report as one station record — which is what a pollution control board expects to receive.

COMMISSIONING & AUDIT

Commissioning covers leak and flow verification, a multipoint linearity check across the working range, and a documented zero and span baseline that becomes the reference for subsequent audits. Thereafter the automatic zero cycle carries day-to-day stability and the scheduled span check confirms it against certified gas, so the record stays defensible without a technician on site.

06 APPLICATIONS

Where it *works.*

Wherever combustion is incomplete, CO has to be measured — for compliance, for traffic and urban air quality reporting, and for understanding what a source actually contributes downwind.

Regulatory CAAQMS Networks	Urban & Traffic Corridors	Thermal Power Stations
Refineries & Petrochemicals	Chemical & Fertiliser Plants	Smelters & Metallurgy
Automotive & Tunnel Testing	EIA & Baseline Studies	Research & Air Quality Studies

WHAT THE DATA SUPPORTS

Continuous CO records serve three audiences at once: regulators, who need hourly and 8-hour values against national standards; operations, who need to see the effect of combustion tuning and load changes; and communities near a corridor or fence line, who need a credible answer about the air they breathe. One measurement, three conversations.

SITING NEAR A SOURCE

Near-road installations are usually placed at breathing height close to congested traffic or a tunnel portal, with a background station away from traffic as reference, so the contribution of vehicles can be separated from what arrives already in the air. Where traffic volume varies sharply, the wider measuring range keeps peak-hour episodes on scale rather than clipped.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Non-dispersive infrared gas filter correlation (NDIR GFC). A curved-mirror long-path cell and a spinning correlation wheel give a self-referencing signal proportional to CO concentration.

Q Which parameter does it measure?

Ambient carbon monoxide concentration, for continuous ambient air quality monitoring in regulatory, industrial and traffic-corridor applications.

Q Is it suitable for continuous monitoring?

Yes. It is designed for real-time, uninterrupted operation, with automatic zero correction so the baseline holds without an operator present.

Q Does it meet CPCB requirements?

The analyzer is designed to meet CPCB and regulatory ambient air quality monitoring guidelines, and reports in the formats a state or central board server expects.

Q Can it be installed outdoors?

Yes, housed in a weather-protected station cabinet. The analyzer itself is a rack instrument; the cabinet provides the environmental protection and shared services.

Q How does the GFC wheel improve accuracy?

It alternates a reference and a measure signal through the same optical path, cancelling slow drift in the source or window that would bias a single-beam reading. The wheel is built for a no-leak, long service life.

Q What ranges are available?

0–50 and 0–100 ppm, selectable by the user from the touchscreen to suit background, urban or near-source conditions.

Q How much data does it hold?

The internal NAND-flash logger stores up to one year of one-minute averaged data, exportable over USB or pulled across the network.

Q How does it integrate with a station?

Ethernet, RS-232, RS-485 and analog outputs let the analyzer report to a station data logger, plant SCADA and a regulatory server simultaneously.

Q What maintenance does it need?

Inlet filter changes, periodic leak and flow checks, and scheduled zero and span verification with certified gas. There are no reagents or wet chemistry.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

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CARBON MONOXIDE, MEASURED CONTINUOUSLY

Know your CO. *In real time.*

Tell us the site, the range you expect and who receives the data, and we'll configure the analyzer and the station around it — calibration, zero air, cabinet and reporting. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a station configuration**
RECOMMENDED
Send your site and reporting requirement — we'll spec ranges, calibration, zero air, cabinet and communications.
sales@vepolink.com →

03 **Book a demonstration**
See a running analyzer and the reporting dashboard with our engineering team.
Request a demo →

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